Haematological Disorders 2: Megaloblastic, Aplastic & Haemolytic Anaemias (HS, G6PD)

Internal Medicine Department (lecturer not named on slides) | Source: `2-_Megaloplastic-aplastic-Hemogobinopathies.pdf` (45 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The lecture has **no answer key** for its case; the answer below is **[reasoning added; key not on slides]**.

⚠️ **Coverage gap:** the file is named "...**Hemoglobinopathies**", but the slides contain **no thalassaemia or sickle-cell content**. Both appear only as names in the haemolytic aetiology list (slide 27). Slide 3's subtopics also omit them. Expect a separate lecture, or ask the lecturer whether they are examinable from this deck. Thalassaemia's iron profile is in **Heme 1, §2.16**.

🔗 **Bridge from earlier lectures:**

- **Heme 1 (CBC & IDA):**
- **macrocytic** branch of the MCV algorithm
- **Howell–Jolly bodies** on the film
- **reticulocytes**: absent in marrow failure, ↑ in haemolysis
- transfusion threshold **Hb < 7**, or with HF/IHD
- **FUO & Travel Medicine Pt 1:**
- **primaquine/tafenoquine haemolysis in G6PD deficiency** ("check G6PD first")
- **malaria** is an infective cause of haemolysis
- ***Salmonella* osteomyelitis in sickle cell**
- **Substance Use Disorders (Khrad): alcohol** → folate deficiency and macrocytosis. **Wernicke** (confusion, nystagmus, ataxia) = **B1**, the lecture-case distractor.
- **Thyroid 1: carbimazole ("Neomercazole")** → agranulocytosis/aplastic anaemia. **Pernicious anaemia** clusters with **autoimmune thyroiditis/Graves**.
- **Gastric Tumours: pernicious anaemia → ↑ gastric carcinoma**; **gastrectomy → B12 deficiency** (loss of intrinsic factor).
- **Lung Cancer (Kotb):** small-cell (oat-cell) carcinoma appears in the slide list of red-cell aplasia causes.
- **Valvular / IE (Reihan): prosthetic and calcific aortic valves** → mechanical (schistocytic) haemolysis.
- **Tuberculosis (Kotb): ileal TB** → B12 malabsorption; **disseminated TB** → marrow failure.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and aetiology of anaemia	Folate trap; MMA vs homocysteine; Heinz bodies
Knowledge	⭐ Manifestations, complications, management	SCD; pernicious anaemia autoimmunity; crises; splenectomy indications
Skill	⭐ Interpret investigations	MCV > 115, hypersegmented PMNs; absent retics; LDH/haptoglobin; osmotic fragility, EMA

Domain	Objective	Likely tested as
Skill	Evidence-based management	B12 IM for life; folate 5 mg; never folate alone in B12 deficiency
Skill	Emergencies	Aplastic infection risk; haemolytic crisis; G6PD trigger avoidance

Main subtopics (slide 3):

- **megaloblastic anaemia** (B12 and folate: pathogenesis, neurology)
- **aplastic anaemia** and the pancytopenia work-up
- **haemolysis basics** (intravascular vs extravascular)
- **G6PD deficiency** (oxidants, Heinz bodies, bite cells)

Hereditary spherocytosis is also taught (slides 36–37).

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2. Main Content (Lecture Order)

2.1 Vitamin B12: Physiology (Slide 5) ⭐⭐

Item	Detail
Source	⭐ **Animal sources only** (not plants)
Requirement	**1 µg/day**
Absorption	Binds **intrinsic factor** (from **gastric parietal cells**) → absorbed in the ⭐ **terminal ileum** via a specific receptor
Transport	**Transcobalamin II**
Stores	⭐ **Liver; enough for 2–3 years** *[added: deficiency takes years to appear]*
Function	**DNA synthesis** in dividing cells; ⭐ **myelination** of nerve fibres

2.2 B12 Deficiency: Aetiology (Slides 6–7) ⭐⭐

1. ↓ Intake:

- dysphagia, starvation
- ⭐ **vegans/vegetarians**

2. ↓ Absorption:

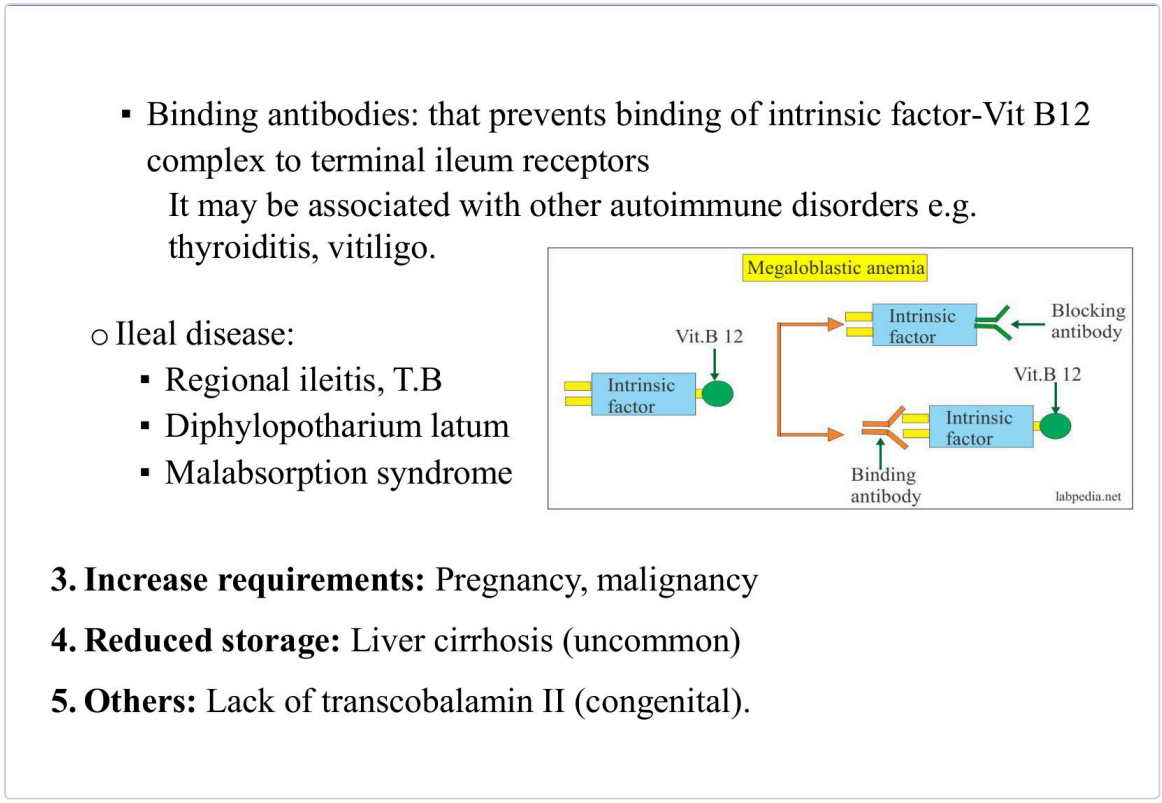
- **gastrectomy, gastritis, gastric carcinoma** (loss of IF)
- ⭐ **Addisonian pernicious anaemia** (autoimmune):
  - classically **northern Europe**; **F > M**; rare **< 30 years**
  - genetic (**HLA-A3, B7**)
  - **three antibody types:**
    1. **anti-parietal cell** antibodies
    2. **blocking** antibodies: stop **B12 binding to IF**
    3. **binding** antibodies: stop the **IF–B12 complex binding the ileal receptor**
  - associated autoimmunity: **thyroiditis, vitiligo** (slide 10 adds **Addison's, Graves'**)
- **ileal disease:**
  - **regional ileitis (Crohn's)**, **ileal TB**
  - ***Diphyllobothrium latum*** (fish tapeworm)

- **malabsorption**
- *[added]* ileal resection, blind-loop bacterial overgrowth, **metformin**, PPIs

3. ↑ Requirement: pregnancy, malignancy.

4. ↓ Storage: liver cirrhosis (uncommon).

5. Other: congenital transcobalamin II deficiency.



Slide 7: Blocking vs binding antibodies in pernicious anaemia

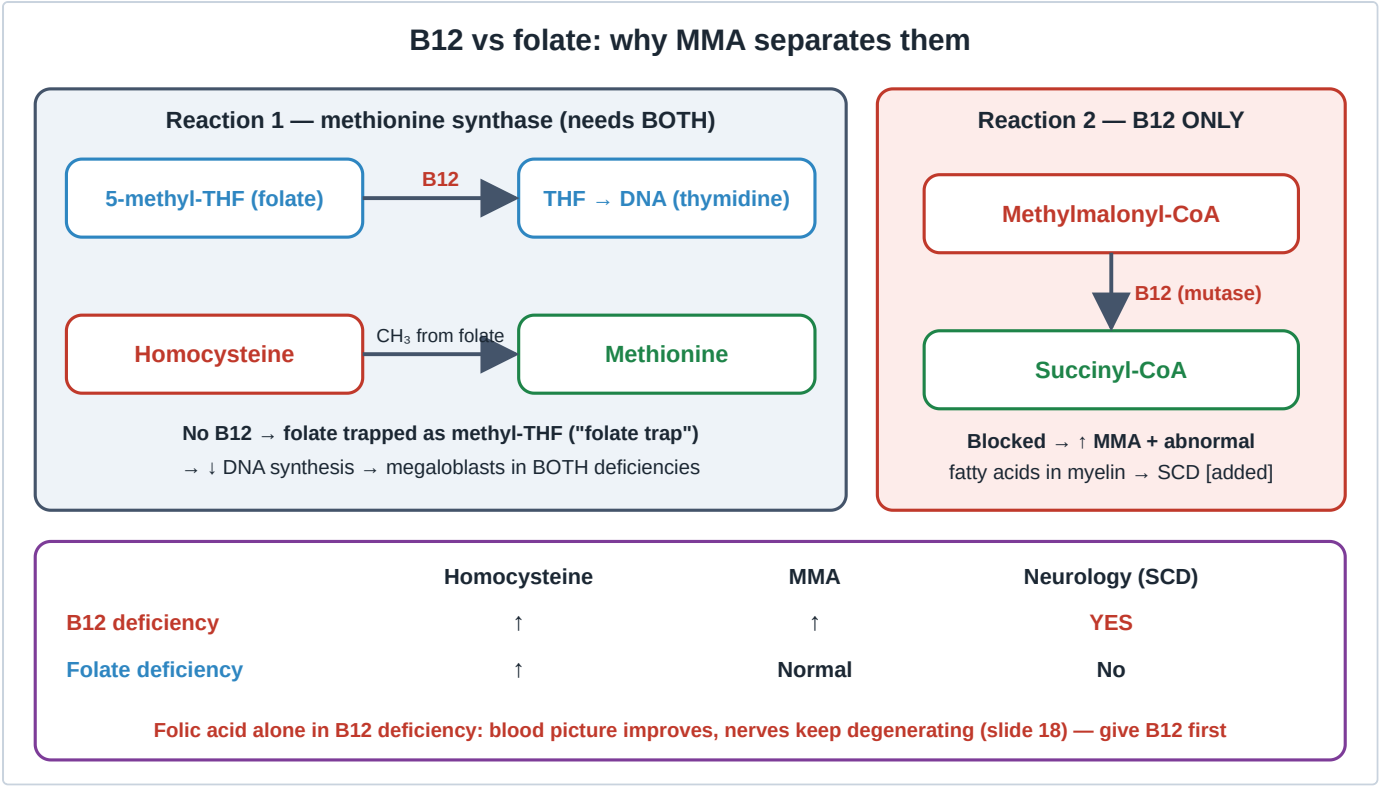
[added] **Note on the antibodies: anti-IF antibodies** (blocking) are the **specific** test for pernicious anaemia (~50 % sensitive, ~100 % specific). Anti-parietal cell antibodies are sensitive but non-specific.

2.3 B12 Deficiency: Pathogenesis (Slides 8–9) ⭐

Mechanism:

- DNA synthesis (B12-dependent) lags behind cytoplasmic growth (RNA unaffected)
- → **nuclear–cytoplasmic asynchrony** → delayed division → ⭐ **megaloblasts**

Site	Consequence
Bone marrow	⭐ **Ineffective erythropoiesis**: megaloblasts die in the marrow → **↑ LDH, ↑ serum iron, ↑ marrow haemosiderin**; platelets destroyed → **thrombocytopenia**; WBCs destroyed → **↑ serum muramidase (lysozyme)**
Peripheral blood	Released cells are destroyed further in the RES → ⭐ **pancytopenia**. RBCs: anisocytosis, poikilocytosis, **Cabot rings, Howell-Jolly bodies**; ↓ platelets; **leucopenia with hypersegmented neutrophils**
GIT	**Epithelial atrophy** (rapid turnover)
CNS	**Defective myelination** → impaired conduction



Added diagram: B12 vs folate in the one-carbon pathway; why MMA separates them

2.4 B12 Deficiency: Clinical Picture (Slide 10) ⭐⭐

1. General anaemia features *[added: often a lemon-yellow tinge from mild haemolysis plus pallor]*.

2. Specific features:

System	Feature
Blood	**Bleeding tendency** (↓ platelets); **infection** (↓ WBCs)
GIT	⭐ **Red glazed tongue** (glossitis); dyspepsia, nausea, vomiting; **diarrhoea**; **hepatosplenomegaly**
CNS	⭐ **Subacute combined degeneration (SCD)** of the cord

Pearls on slide 10:

- SCD affects the **spinocerebellar tracts, lateral corticospinal tracts and dorsal columns**
- ⭐ **Always consider B12 deficiency when evaluating dementia**

[added] **SCD clinically:**

- **dorsal columns:** ↓ vibration and proprioception, **+ Romberg**, sensory ataxia
- **corticospinal:** spastic weakness, **extensor plantars**
- combined with **peripheral neuropathy: absent ankle jerks**
- cognitive change ("megaloblastic madness")
- neurology can occur **without anaemia**

3. Features of the cause (pernicious anaemia):

- other autoimmune disease: Addison's, vitiligo, Graves'
- ⭐ **↑ gastric carcinoma** risk

2.5 B12 Deficiency: Investigations (Slides 11–13) ⭐⭐⭐

Investigations

I. ↓ Hb, ↓ PCV, ↓ RBCs, ↑ESR.

II. Type of anemia: Megaloblastic normochromic:
- MCV(> 115 FL). - Normal MCHC.

III. The cause:

➢ **Pancytopenia**

1. **RBCs:** Howel-Jolly bodies, cabbot rings, anisocytosis & poikilocytosis +
   Platelets: Thrombocytopenia + • bleeding time.
2. **WBCs:** Leucopenia + • segmented cells + • muramidase.
3. **BM:** -Megaloblast precursors. - Reticulocytes (characterstic).

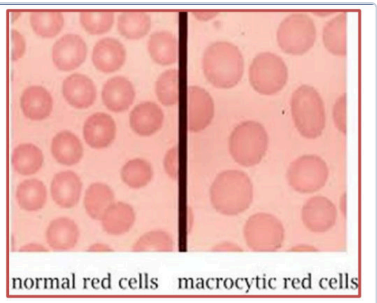
➢↓ **Vit B12**

1. Low serum Vit B12 (200 – 800 Pg/ml).
2. IM Vit B12 → Reticulocytosis (therapeutic test).

➢ **Methylmalonic acid (MMA):**

↑ Homocysteine + ↑ MMA → B12 deficiency

↑ Homocysteine + normal MMA → Folate deficiency



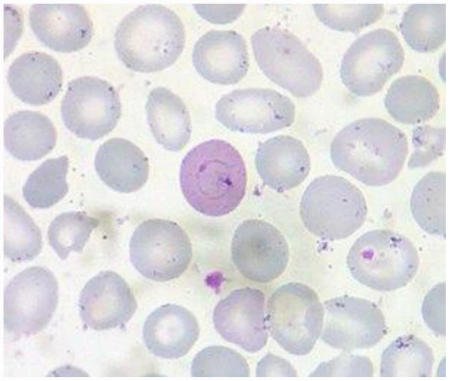
Slide 11: Normal vs macrocytic red cells

1. ↓ Hb, ↓ PCV, ↓ RBCs, ↑ ESR
2. **Type: megaloblastic, normochromic**: ⭐ **MCV > 115 fL**, **normal MCHC**
3. **Cause:**

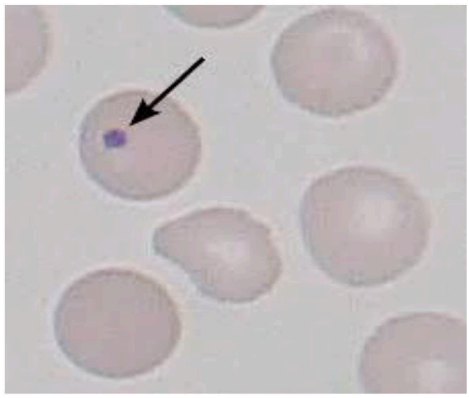
Domain	Findings
Pancytopenia	RBCs: **Howell-Jolly bodies, Cabot rings**, aniso/poikilocytosis · platelets: thrombocytopenia + ↑ bleeding time · WBCs: leucopenia + ⭐ **hypersegmented neutrophils** (*[added]* ≥ 6 lobes, or > 5 % with 5 lobes) + ↑ muramidase
Bone marrow	**Megaloblastic precursors**; reticulocytes (the slide says "characteristic"; see note)
↓ B12	Low serum B12 (slide: "200–800 pg/mL", which is actually the **normal** range); ⭐ **IM B12 → reticulocytosis** (therapeutic test)
⭐ **MMA & homocysteine**	**↑ homocysteine + ↑ MMA → B12 deficiency** · **↑ homocysteine + normal MMA → folate deficiency**

⚠️ **Slide 11 issues:**

- **"Low serum B12 (200–800 pg/mL)"**: 200–800 is the **normal range**. Deficiency is **< 200 pg/mL** *[added]*; 200–300 is borderline, so check MMA.
- **"BM: Reticulocytes (characteristic)"**: an arrow was probably lost. Megaloblastic anaemia has a **low reticulocyte count** (ineffective erythropoiesis). The rise after IM B12 is the therapeutic test.
- The "•" symbols in "• bleeding time", "• segmented cells" and "• muramidase" are lost ↑ arrows.



Cabot rings are thin, red-violet staining, threadlike strands in the shape of a loop or figure-8 that are found on rare occasions in red blood cells (erythrocytes)



- Howell–Jolly bodies are histopathological findings of basophilic nuclear remnants (clusters of DNA) in circulating erythrocytes. During maturation in the bone marrow late erythroblasts normally expel their nuclei, but in some cases a small portion of DNA remains

Slide 12: Cabot ring and Howell–Jolly body

- **Howell–Jolly bodies:** basophilic **nuclear remnants (DNA)** left when late erythroblasts don't fully expel the nucleus.
- **Cabot rings:** thin red-violet **loop or figure-8 strands** in RBCs.

4. Aetiology (slide 13):

- **Pernicious anaemia:**
  - **antibodies**
  - **gastric function test for achlorhydria**
  - ⭐ **endoscopy to exclude gastric carcinoma**
- **Malabsorption work-up**
- **Schilling test** (*[added: historical, no longer available]*):
  - **procedure: 1000 µg B12 IM** (saturates binding sites) → then **1 µg oral radiolabelled B12**
  - **normal: ≥ 25 %** of the label appears in 24 h urine
  - **abnormal: < 5 %** → malabsorption

VI. Investigation of Etiology: e.g.

➢**Preniciouse aneamia:**

- Detection of antibodies.
- Gastric function test for achlorhydria.
- Endoscopy to exclude gastric carcinoma.

➢**Investigations for malabsorption syndrome.**

- **Shilling test:**
  - **Procedure**: 1000 ug B12 IM → Then 1 Ug oral radioactive B12.
  - **Normally**: 25% of radioactive Vit B12 appears in urine / 24 hrs
  - **Abnormal response**: < 5% of radioactive B12 / 24 hr urine → Malabsorption.

	Description	Radiolabeled vitamin B_{12} levels in urine
Stage 1	• Oral radiolabeled vitamin B_{12} PLUS IM dose of unlabeled vitamin B_{12}	• Normal excretion: Insufficient dietary intake or an increased demand is the most likely cause. • Excretion below what is expected: Perform stage 2.
Stage 2	• Oral radiolabeled vitamin B_{12} PLUS intrinsic factor	• Normal excretion: Malabsorption due to lack of intrinsic factor is the most likely cause. • Excretion remains below expected: Perform stage 3.
Stage 3	• Oral radiolabeled vitamin B_{12} PLUS antibiotics	• Normal excretion indicates bacterial overgrowth. • Excretion remains below expected: Perform stage 4.
Stage 4	• Oral radiolabeled vitamin B_{12} PLUS pancreatic enzymes	• Normal excretion indicates pancreatic enzyme deficiency (e.g., chronic pancreatitis).

Slide 13: Schilling test stages

Stage	Oral radiolabelled B12 plus…	Normal excretion means
1	IM unlabelled B12	**Dietary** deficiency / ↑ demand (if low → stage 2)
2	**Intrinsic factor**	⭐ **Pernicious anaemia** (IF lack) (if low → stage 3)
3	**Antibiotics**	**Bacterial overgrowth** (if low → stage 4)
4	**Pancreatic enzymes**	**Pancreatic insufficiency** (chronic pancreatitis)

2.6 B12 Deficiency: Treatment (Slide 14) ⭐

I. Replacement:

- **hydroxocobalamin 100 µg IM daily** until RBC count reaches **5 million/mm³**
- then **one ampoule IM monthly for life** in malabsorption (pernicious anaemia)

II. Treat the cause.

III. Transfuse if **Hb < 7** or **HF/IHD**.

[added] **Notes:**

- UK practice is **1 mg IM** (3×/week for 2 weeks, then every 2–3 months); high-dose **oral** B12 works even in pernicious anaemia
- check **K^+** early: hypokalaemia can occur as new cells take up K^+
- iron deficiency may be unmasked as cells regenerate

🔁 **Recap: B12.**

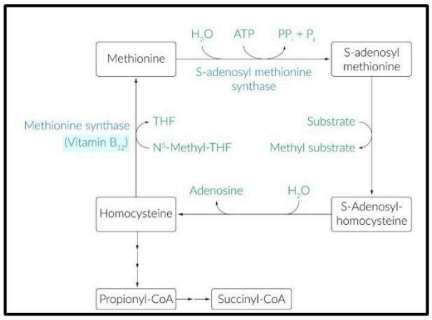
- **Physiology:** animal foods only; **IF + terminal ileum**; 2–3-year liver stores.
- **Causes: pernicious anaemia** (3 antibodies; ↑ gastric ca), gastrectomy, ileal disease, fish tapeworm, vegans.
- **Clinical:** pancytopenia, glossitis, **SCD** (dorsal columns, corticospinal, spinocerebellar), dementia.
- **Labs: MCV > 115**, **hypersegmented PMNs**, **↑ LDH**, **↑ MMA + ↑ homocysteine**.
- **Treatment:** IM hydroxocobalamin, lifelong in PA.

🔗 **Link:** folate gives the **same blood picture** (the methionine-synthase step needs both), but **no neurology and normal MMA**.

▶️ **Up next: Folate deficiency.**

2.7 Folate Deficiency (Slides 15–18) ⭐⭐

Physiology



- **Source:** In vegetables.
- **Requirements:** 50 ug / day.
- **Absorption:** Intestine.
- **Stores**: In liver 5-15 mg (sufficient for only 2-3 months).
- **Function:** For DNA synthesis.

➢**Vit B12 and folic acid are needed for methylation of homocystiene to methionine which is needed for conversion of uracil to thymidine to be incorporated in DNA**

Slide 15: Methionine cycle (B12 and folate)

Item	Folate (vs B12)
Source	**Vegetables** (B12: animal)
Requirement	**50 µg/day** (B12: 1 µg)
Absorption	Intestine (*[added]* jejunum) (B12: terminal ileum + IF)
Stores	Liver **5–15 mg**: ⭐ **only 2–3 months** (B12: 2–3 years)
Function	DNA synthesis

Why both vitamins matter: B12 and folate are both needed to **methylate homocysteine → methionine**, which is needed to convert **uracil → thymidine** for DNA.

Aetiology:

1. ↓ **Intake:** anorexia, ⭐ **alcoholism**, low vegetable intake

2. ↓ **Absorption:** malabsorption; drugs: **anticonvulsants** (*[added]* phenytoin), methotrexate
3. ↓ **Storage:** liver cirrhosis
4. ⭐ ↑ **Requirement: pregnancy**, malignancy, **haemodialysis**, **haemolytic anaemia**
5. ↑ **Utilisation / antagonism: ⭐ methotrexate, trimethoprim** (DHFR inhibitors)

Clinical: as B12, **except no CNS manifestations**.

Investigations:

- ↓ Hb, PCV, RBCs; ↑ ESR; type as B12
- **low serum folate** (N **6–20 ng/mL**)
- ↑ **homocysteine** (normal MMA)
- **FIGLU test** (historical): histidine → FIGLU → glutamate; in folate deficiency **urinary FIGLU ↑**
- *[added]* **red-cell folate** reflects stores better than serum folate

Treatment:

1. **Folic acid 5–15 mg orally daily** (*[added]* usually 5 mg daily for 4 months)
2. Treat the cause (e.g. stop methotrexate; *[added]* folinic acid rescue)
3. Transfusion if needed

Student pearl: folic acid supplementation for anyone **planning or capable of pregnancy** → prevents **neural tube defects**.

⭐⭐ **NB (slide 18): empirical folic acid in B12 deficiency worsens the CNS disease**, because it consumes the already-low B12. *[added]* Folate corrects the blood picture while the neuropathy progresses. **Always give B12 first, or both, until B12 deficiency is excluded.**

> 🔁 **Recap: Folate.**
> - **Physiology:** vegetables; **3-month stores**.
> - **Causes:** alcohol, pregnancy, haemolysis, dialysis, **methotrexate/trimethoprim/anticonvulsants**.
> - **Clinical:** same blood picture as B12; **no neurology**; **MMA normal**.
> - **Treatment: 5 mg folic acid**; never give folate alone if B12 might be low.
>
> 🔗 **Link:** megaloblastic anaemia causes **pancytopenia with a cellular marrow**. Aplastic anaemia causes **pancytopenia with an empty marrow**; they sit in the same pancytopenia DD.
>
> ▶️ **Up next: Aplastic anaemia.**

2.8 Bone-Marrow Failure (Aplastic Anaemia): Types & Aetiology (Slides 20–21) ⭐⭐

Types (slide 20):

Type	Examples
Unipotent: red cells	↓ **EPO: renal failure**; **anti-EPO / red-cell aplasia: oat-cell (small-cell) carcinoma, thymoma** (slide); ⭐ **aplastic crisis of haemolytic anaemias** (parvovirus)
Unipotent: platelets	⭐ **TAR syndrome**: **T**hrombocytopenia + **A**bsent **R**adii
Unipotent: WBCs	Idiopathic, congenital
⭐ **Pluripotent**	Stem-cell failure → **pancytopenia: empty marrow and empty blood**

> *[added]* **Thymoma** → **pure red cell aplasia** is the classic association (autoimmune). Small-cell carcinoma is a less typical example.

Aetiology of pluripotent failure (slide 21):

1. ⭐ **Idiopathic (50 %)**: **autoimmune cytotoxic T-cell** suppression of the marrow
2. **Irradiation** (e.g. radioactive phosphorus)
3. ⭐ **Drugs ("anti's"):**
   - **antithyroid: Neomercazole (carbimazole)**
   - "anti-emetic": **chlorpropamide**
   - **antibiotic: chloramphenicol**
4. **Chemicals: benzene**, insecticides
5. **Infections:** disseminated TB, **EBV, hepatitis** (*[added]* seronegative hepatitis; **parvovirus B19** = red-cell aplasia)
6. **Immune:** SLE
7. **Marrow replacement:** leukaemia, lymphoma (**myelophthisic anaemia**)

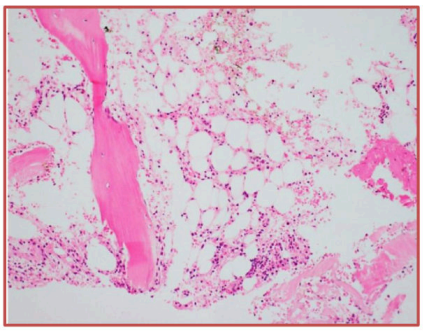
> ⚠️ **Slide error:** chlorpropamide is listed as an "**anti-emetic**". It is a **sulfonylurea (antidiabetic)**. *[added]* Other classic drug causes: **carbamazepine, phenytoin, gold, NSAIDs (phenylbutazone), sulfonamides**, cytotoxics.

2.9 Aplastic Anaemia: Clinical, Investigations, Treatment (Slides 22–24) ⭐⭐

Clinical:

1. General anaemia features
2. ↓ **Platelets:** purpura, bleeding
3. ↓ **WBCs:** infection, especially **fungal/opportunistic**. Infection affects sites already colonised by bacteria (e.g. mouth), and **there is no pus** (no neutrophils).
4. Features of the cause, e.g. **skeletal deformities** (*[added]* Fanconi anaemia: short stature, thumb/radial anomalies, café-au-lait spots)
5. *[added]* **No splenomegaly**. Splenomegaly suggests another diagnosis (hypersplenism, leukaemia, myelofibrosis).

Investigation



1. **For all anemia:** ¯ Hb, HCT, RBCs & • ESR
2. **Type of anemia:** Normocytic normochromic
   - MCV, MCH normal.
3. **Causes**:
   - ↓platelets & WBCs.
   - Absent reticulocytes (most specific).
4. **Aetiology:** Bone narrow biopsy shows hypoplastic narrow

Differential diagnosis: Causes of pancytopenia:

- Hypersplenism
- Megaloplastic anemia
- Aplastic anemia

Slide 23: Hypocellular (fatty) marrow biopsy

Investigations:

1. ↓ Hb, Hct, RBCs; ↑ ESR
2. **Type: normocytic normochromic** (*[added]* often mildly macrocytic)
3. ↓ platelets and WBCs; ⭐ **absent reticulocytes (most specific)**
4. ⭐ **Bone-marrow biopsy: hypoplastic (fatty) marrow**. *[added]* Trephine biopsy, not aspirate (aspirate gives a "dry tap").

DD of pancytopenia (slide 23):

- **hypersplenism**
- **megaloblastic anaemia**
- **aplastic anaemia**
- *[added]* also marrow infiltration (leukaemia, myelofibrosis), PNH, SLE, sepsis

Treatment (slide 24):

Category	Measures
1. Supportive	**Packed RBCs**; **platelets**; **avoid trauma** (shaving, **IM injections**); antibiotics (**imipenem**) + **antifungal + antiviral**
2. Marrow stimulation	**Steroids**; **androgens** (↑ RBCs); **GM-CSF** (↑ WBCs); **lithium**
3. Treat the cause	Idiopathic: **immunosuppression** (slide: prednisone); **stop the drug**
4. ⭐ Bone-marrow transplantation	(definitive)

⚠️ **Outdated on slide** *[added]*: current first line for severe aplastic anaemia is:

- **HSCT** (young patient with a matched sibling)
- otherwise **immunosuppression with ATG + ciclosporin + eltrombopag**

Steroids alone, lithium and androgens are historical or adjunct. Use irradiated, leucodepleted blood products, and limit transfusions in transplant candidates.

🔁 **Recap: Aplastic anaemia.**

- **Types:** unipotent (red-cell aplasia, TAR, neutropenia) vs pluripotent (pancytopenia).
- **Causes: 50 % idiopathic (T-cell)**, **carbimazole/chloramphenicol**, benzene, radiation, hepatitis/EBV, SLE.
- **Findings:** normocytic, **absent retics**, **hypocellular trephine**; no splenomegaly *[added]*.
- **Treatment:** supportive (no IM injections) → immunosuppression / **BMT**.

🔗 **Link:** aplasia = **production failure, no retics**. Haemolysis is the opposite: **destruction with high retics**.

▶️ **Up next: Haemolytic anaemia.**

2.10 Haemolytic Anaemia: Physiology & Aetiology (Slides 26–28) ⭐⭐

Physiology (slide 26):

- **Definition: shortened RBC life span**; normal = ⭐ **120 days**.
- The marrow compensates (up to 6–8× output). Anaemia appears when life span falls to **1/6–1/8 of normal (15–20 days)** = **uncompensated** haemolysis.
- ⭐ **Site: 90 % extravascular** (RES: spleen, liver) and **10 % intravascular**.

Causes (slides 27–28):

1. Corpuscular (intrinsic)	2. Extracorpuscular (extrinsic)
A. Membrane: hereditary spherocytosis, hereditary elliptocytosis, **PNH**	**A. Mechanical: prosthetic or sclero-calcific aortic valve**; **MAHA**: **DIC, HUS, malignant HTN, scleroderma, TTP**
B. Enzyme: ⭐ G6PD deficiency, **pyruvate kinase deficiency**	**B. Hypersplenism**
C. Haemoglobin: thalassaemia, sickle cell (not taught here)	**C. Immune: autoimmune** (**warm or cold** AIHA); **alloimmune** (**incompatible transfusion, Rh incompatibility**)
	D. Infective: sepsis, **malaria**, haemolytic streptococci (*[added] Clostridium perfringens*)
	E. Toxic: snake/spider venom, lead, staphylococcus, copper (Wilson)

⚠️ **Slide 28 layout error: TTP** is printed as item 5 under the *immunological* heading, where it breaks into the numbering. It belongs with **MAHA** (mechanical). Slide 28 also numbers MAHA causes 1–4 and continues to 5 under the next heading.

2.11 Haemolytic Anaemia: Clinical Picture (Slides 29–32) ⭐⭐

I. Anaemia features (when uncompensated).

II. Extravascular haemolysis:

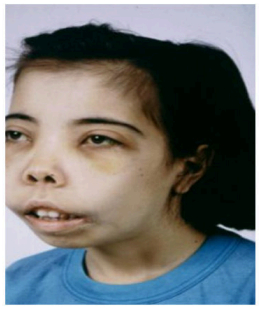
- **Bone: expansion of bone ends** → **mongoloid facies** (frontal bossing, maxillary overgrowth) (*[added]* chronic childhood haemolysis, thalassaemia major)
- **Hepatosplenomegaly:** extramedullary haematopoiesis + haemolysis + haemosiderosis
- **Jaundice:**
  - **acholuric/indirect**
  - the liver can raise conjugation **~8×**, so jaundice appears only with severe haemolysis
  - ⭐ **pigment gallstones**
- ⭐ **Haemosiderosis** (transfusional/haemolytic iron overload):
  - cardiomyopathy, **"bronze diabetes"**, skin ulcers
  - interstitial lung fibrosis, **testicular atrophy**, renal tubular defects
- **Dark stool**; **urine darkens on standing** (urobilinogen → urobilin)

Clinical Picture

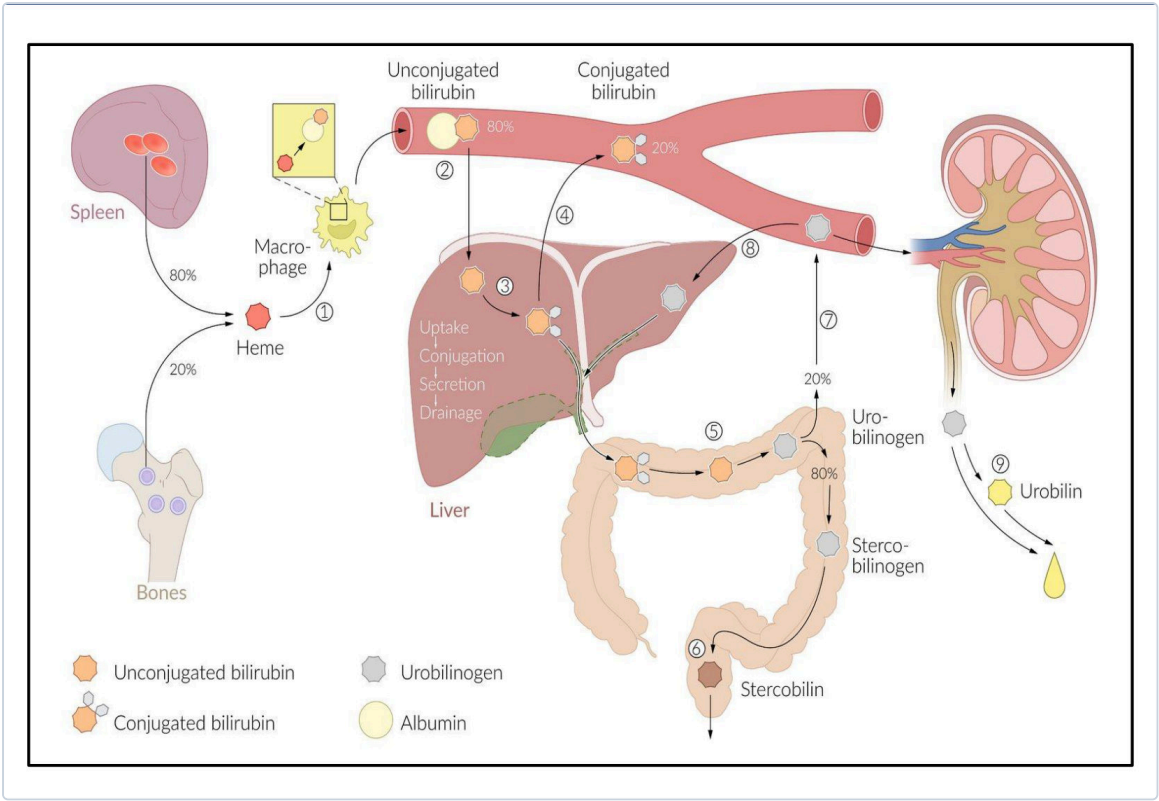
I. Of anemia : when hemolysis is uncompensated.

II. Of hemolytic anemia.

- **Extravascular hemolysis:**
  - Bone: Expansion of bone ends leading to Mongoloid features.
  - Hepatosplenomegaly: Due to extramedullary hematopoiesis, & hemolysis, hemosidrosis.
  - Jaundice (liver can increase conjugation 8 times and jaundice will occur only in severe hemolysis) & gall bladder stones.
  - Hemosedrosis:
    - Cardiomyopathy.
    - Bronze diabetes.
    - Skin ulcers.
    - Interstitial lung fibrosis.
    - Testicular atrophy.
    - Renal tubular defects.
  - Dark stool and urine darkens on standing.



Slide 29: Haemolytic facies



Slide 30: Bilirubin metabolism (unconjugated → conjugated → urobilinogen/stercobilin)

Crises (slide 31):

Crisis	Cause	Pallor	Jaundice/bilirubin	Retics
Haemolytic	↑ haemolysis (infection, drugs)	↑	⭐ ↑	↑
⭐ **Aplastic**	⭐ **Parvovirus B19**	↑	**No change**	*[added]* **↓↓ (absent)**
Megaloblastic	**Folate (and B12) depletion** from ↑ demand	↑	—	↓ + **pancytopenia**, neuro/GI features

[added] **Sequestration crisis** (sickle cell: spleen pools blood, Hb falls with splenomegaly) is not on the slides.

III. Intravascular haemolysis (slide 32):

- free Hb in plasma → filtered by the kidney → ⭐ **haemoglobinaemia + haemoglobinuria**
- this happens **after the binding proteins are exhausted**: **haptoglobin**, **haemopexin** (research use), albumin
- *[added]* also **haemosiderinuria** (days later), dark urine, and AKI in massive haemolysis

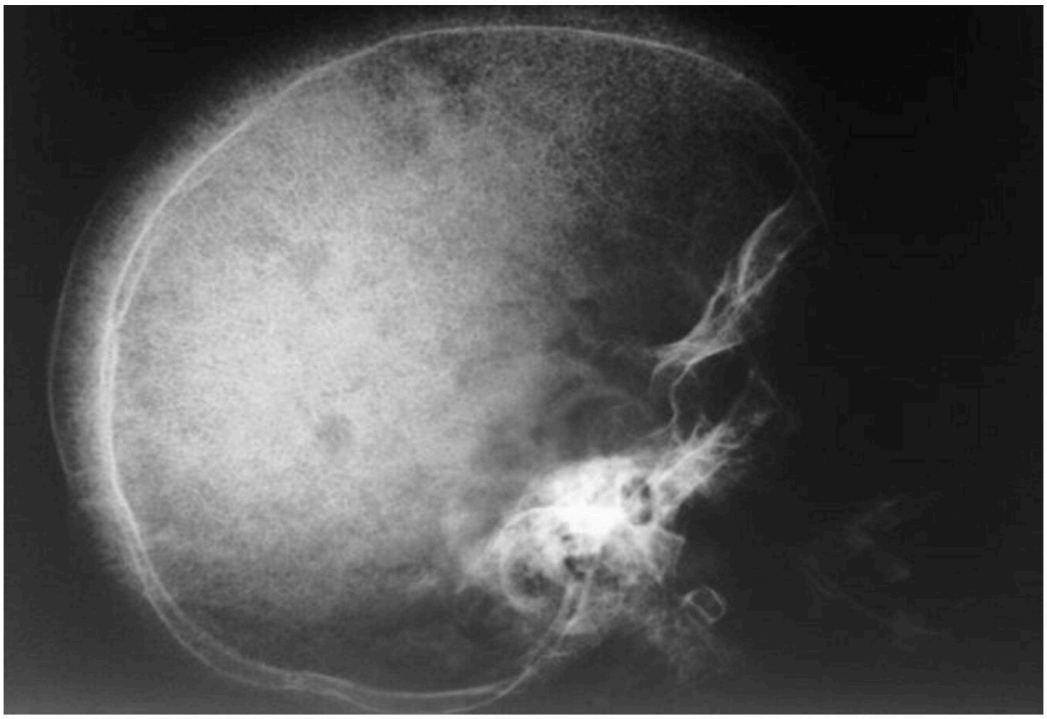
2.12 Haemolytic Anaemia: Investigations & Treatment (Slides 33–35) ⭐⭐

1. ↓ Hb, RBCs, Hct; ↑ ESR
2. **Type: mostly normocytic normochromic** (*[added]* MCV may rise with reticulocytosis)
3. **Haemolysis markers:**

Test	Finding
RBC survival	Shortened (**^{99m}Tc / ^{51}Cr-labelled RBCs**)
⭐ **LDH**	**↑**
Skull X-ray	Marrow expansion: widened medulla, thin cortex, ⭐ **"hair-on-end"**
⭐ **Bilirubin**	**↑ total, mainly indirect**; **↑ urine urobilinogen**, ↑ faecal stercobilinogen

Test	Finding
Iron	↑ serum iron and ferritin, ↓ TIBC
Ultrasound	Hepatosplenomegaly, **gallstones**
WBCs & platelets	↑ (hyperactive marrow)
⭐ **Reticulocytes**	**↑ (reticulocytosis)**
Intravascular	⭐ **↓ haptoglobin**, ↓ haemopexin

[added] Also order a **direct antiglobulin (Coombs) test** to separate immune from non-immune haemolysis, and a **blood film** (spherocytes, schistocytes, bite cells).



Slide 34: Skull X-ray, "hair-on-end" appearance

⭐ **Clinical pearl (slide 35): haptoglobin is low in BOTH intravascular and extravascular haemolysis**, so it **should not be used to tell them apart**. Haemoglobinuria and haemosiderinuria point to intravascular haemolysis *[added]*.

Treatment (slide 35):

1. **Blood transfusion**
2. **Folic acid** (and B12) replacement (↑ demand, to prevent megaloblastic crisis)
3. ⭐ **Iron chelation: desferrioxamine SC** (*[added]* oral deferasirox, deferiprone)
4. **Splenectomy** (selected)
5. Treat the cause

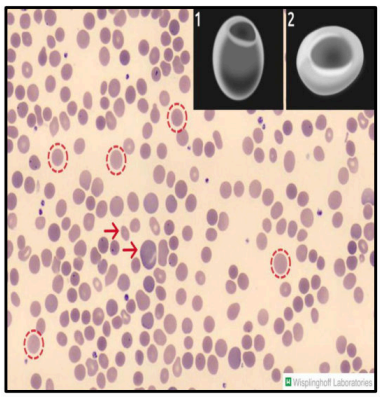
2.13 Hereditary Spherocytosis (Slides 36–37) ⭐⭐

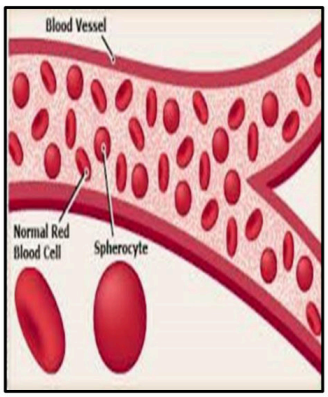
Etiology:

- Hereditary (autosomal dominant)
- There is a defect in the protein composition in RBCs cell wall (ankyrin & spectrin) that makes them more liable to hemolysis in the RES& leads to reduction of RBCs size which become small and spherical in shape(microspherocytes) making them more liable to osmotic rupture.

Clinical Picture:

1. As anemia in general
2. As hemolytic anemia
3. Positive family history





Slide 36: Spherocytes on film; splenic trapping

Aetiology & pathogenesis:

- **autosomal dominant** (*[added]* ~75 %)
- ⭐ defect in membrane proteins **ankyrin and spectrin** (*[added]* also band 3, protein 4.2)
- RBCs lose membrane → **small spherical cells (microspherocytes)** → trapped and destroyed in the **spleen (RES)**, and **osmotically fragile**

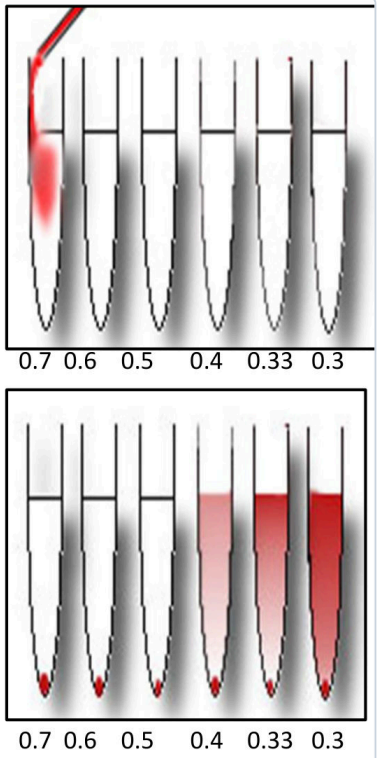
Clinical: general anaemia + haemolytic features + ⭐ **positive family history** (*[added]* jaundice, splenomegaly, gallstones, parvovirus aplastic crisis).

Investigations:

1. Anaemia and haemolysis markers
2. **Type:** "**microcytic hyperchromic**" (*[added]* MCV often normal; the defining feature is ⭐ **↑ MCHC**)
3. **Specific: spherocytes** on the film (no central pallor)
4. **Osmotic fragility test:**
   - RBCs placed in graded NaCl dilutions
   - **normal:** haemolysis **starts at 0.4 % and is complete at 0.33 %**
   - **HS:** haemolysis **starts at 0.6 %** (↑ fragility)
5. ⭐ **EMA binding test** (eosin-5′-maleimide flow cytometry): **most sensitive and specific**
6. *[added]* **DAT negative**, which separates HS from warm AIHA (also spherocytic)

Investigations:

1. For anemia & hemolytic anemia.
2. Type: micorcytic hyperchromic
3. Specific: spherocytes
4. Of cause: Osmotic fragility test: adding RBCs to different dilutions of NaCl (normally hemolysis starts at 0.4 and completed at 0.33, but in spherocytosis it starts at 0.6).
5. EMA binding test (Eosin-5'-maleimide flow cytometry) It is the most sensitive and specific



Treatment

1. As general
2. Splenectomy(better after age of 4) in patients with:
   - High transfusion requirements
   - Family history of death due to sherocytosis
   - Hypersplenism

Slide 37: Osmotic fragility (HS haemolyses at higher saline concentrations)

Treatment:

1. General (folic acid, transfusion as needed)
2. ⭐ **Splenectomy** (**better after age 4**) for:
   - **high transfusion requirement**
   - **family history of death from spherocytosis**
   - **hypersplenism**
3. *[added]* Before splenectomy, **vaccinate** (pneumococcus, meningococcus, Hib). Afterwards, **Howell-Jolly bodies** appear on the film (link to Heme 1). Consider cholecystectomy if gallstones are present.

2.14 G6PD Deficiency (Slides 38–40) ⭐⭐⭐

Inheritance: ⭐ **X-linked**: **males affected**, **females carriers**.

Pathogenesis (slide 38):

1. **G6PD** is the **first step of the hexose monophosphate (pentose phosphate) shunt**
2. the shunt makes **NADPH**, which regenerates **reduced glutathione** from oxidised glutathione
3. reduced glutathione is the **first defence against oxidation of Hb to methaemoglobin**
4. ↓ G6PD → oxidation of membrane **SH groups** and **Hb denaturation** → Hb precipitates on the membrane = ⭐ **Heinz bodies** → cells removed in the **RES** (*[added]* splenic macrophages "bite" them out → ⭐ **bite cells**)

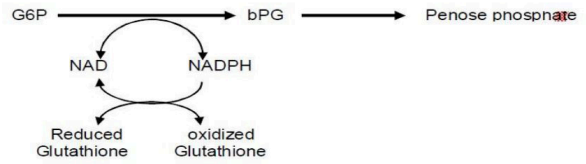
Etiology:

X-linked disease:

- Affect males.
- Females are carriers.

Pathogenesis:

- G6PD enzyme is the 1st step in hexose monophosphate shunt (HMP).
- The shunt is responsible for regenerating reduced Glutathione from oxidized Glutathione by NADPH (H+ donor).
- Reduced glutathione is the 1st protective line against oxidation of Hb to met HB.
- ↓G6PD → oxidation of SH groups in RBCs membrane
  → denaturation of Hb which attaches to RBCs cell membrane (Heinz bodies)
  → destruction of these RBCs in RES



Slide 38: HMP shunt: G6PD → NADPH → reduced glutathione

Clinical picture (slide 39): haemolysis triggered by oxidants:

1. ⭐ **Infection** (commonest trigger *[added]*)
2. ⭐ **Drugs:**
   - **antimalarials** (*[added]* **primaquine**, tafenoquine, dapsone)
   - **anti-TB drugs**
   - **sulpha drugs**
   - **antipyretics: aspirin** (high dose)
   - **synthetic vitamin K**
   - *[added]* nitrofurantoin, rasburicase, methylene blue
3. **Metabolic acidosis** (e.g. **DKA**)
4. ⭐ **Uncooked fava beans (favism)**

Variants and favism (slide 39):

Variant	Defect	Severity
G6PD A⁻ ("African"; slide: "negros")	Enzyme deficient in **ageing RBCs only**	**Mild**, self-limiting
⭐ **G6PD Mediterranean**	Deficient in **young RBCs too**	⭐ **Severe haemolysis** (*[added]* common in Saudi Arabia / Middle East)

Favism (slide's model): needs **two defective genes**: the **G6PD gene**, plus a second gene defect that releases **O_2 free radicals** when fava beans are metabolised.

Investigations (slide 40):

1. Anaemia and haemolysis markers (*[added]* largely **intravascular** → haemoglobinuria, dark urine)
2. **Specific:**
   - ⭐ **Heinz bodies** during the attack, stained with **methyl violet** (supravital; *[added]* invisible on routine Romanowsky stain, which shows **bite/blister cells**)

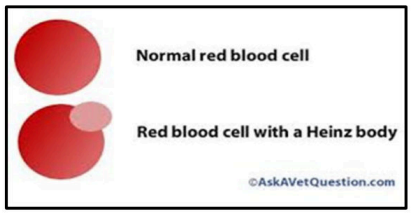
- ⭐ **↓ G6PD activity**, measured **2 months after the last haemolytic attack**. Young RBCs made during the crisis have **normal G6PD levels** → **false-normal** result during or just after the attack.

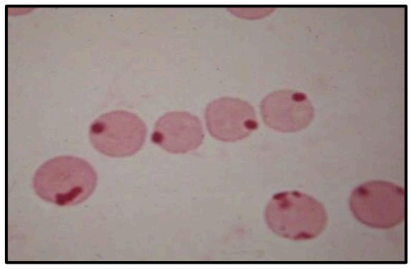
Investigations:

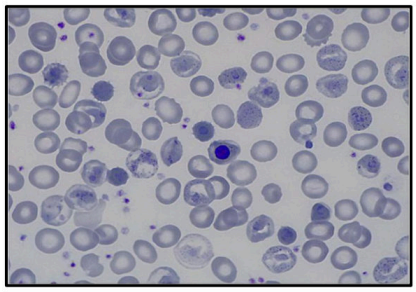
1. As anemia: ↓ Hb, PCV & RBCs, ↑ ESR.
2. As hemolytic anemia
3. Specific
   - Heinz bodies seen during the attack (stained with methyl violet).
   - ↓ G6PD activity in RBCs: done only 2 months after the last haemolytic attack as the newly generated RBCs during crises have normal level of G6PD.

Treatment:

- In crisis → blood transfusion.
- A void cause → treatment of infection & stop causative drug







Slide 40: Heinz bodies and bite cells

Treatment:

- **crisis: blood transfusion**
- ⭐ **avoid the cause**: treat infection, stop the drug, avoid fava beans
- *[added]* hydration; counsel the patient; check G6PD before primaquine/dapsone/rasburicase

> 🔁 **Recap: Haemolysis.**
> - **Basics:** RBC life 120 d; anaemia when < 15–20 d; **90 % extravascular**.
> - **Causes:** intrinsic (membrane/enzyme/Hb) vs extrinsic (mechanical/MAHA, hypersplenism, immune, infective, toxic).
> - **Markers: ↑ LDH, ↑ indirect bilirubin, ↑ retics, ↓ haptoglobin** (low in both types), hair-on-end skull, gallstones.
> - **Crises: aplastic = parvovirus, no ↑ bilirubin**.
> - **HS:** AD, spectrin/ankyrin, **↑ osmotic fragility (starts 0.6 %)**, **EMA** best, **splenectomy after age 4**.
> - **G6PD:** X-linked, HMP shunt/NADPH/glutathione → **Heinz bodies, bite cells**; triggers = infection, primaquine, sulpha, **fava**, DKA; **assay 2 months after the attack**.

2.15 Case & Summary (Slides 41–43)

See Q1 in §7.

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3. High-Yield Comparison Tables

Table A: B12 vs Folate Deficiency

[slides 5–18 + added]

	Vitamin B12	Folate
Source	**Animal only**	**Vegetables**
Daily need	**1 µg**	**50 µg**
Absorption	**IF + terminal ileum**	Jejunum *[added]*
Stores	**2–3 years**	**2–3 months**
Key causes	**Pernicious anaemia**, gastrectomy, ileal disease (Crohn's, TB), *D. latum*, vegans, *[added]* metformin	**Alcohol**, poor diet, **pregnancy**, haemolysis, dialysis, **methotrexate, trimethoprim, anticonvulsants**
Blood picture	Megaloblastic, MCV > 115, **hypersegmented PMNs**, pancytopenia	Same
Neurology (SCD)	⭐ **Yes**	❌ **No**
Homocysteine	↑	↑
⭐ **MMA**	**↑**	**Normal**
Special tests	Antibodies (anti-IF, anti-parietal), Schilling (historical)	FIGLU (historical), red-cell folate
Treatment	**Hydroxocobalamin IM** (lifelong in PA)	**Folic acid 5–15 mg PO**
Trap	—	⚠️ **Folate alone in B12 deficiency worsens neurology**

Table B: Pancytopenia DD

[slide 23 + added]

	Megaloblastic	Aplastic	Hypersplenism
MCV	**↑↑ (> 115)**	N (or mildly ↑)	N
Retics	↓	⭐ **Absent**	N/↑
Marrow	**Hypercellular, megaloblastic**	⭐ **Hypocellular, fatty**	Hypercellular
Spleen	± mild	**Not enlarged** *[added]*	⭐ **Enlarged**
Film	Hypersegmented PMNs, Howell–Jolly, Cabot rings	Unremarkable	Unremarkable

Table C: Intravascular vs Extravascular Haemolysis

[slides 26, 29, 32–35 + added]

	Extravascular (90 %)	Intravascular (10 %)
Site	Spleen/liver macrophages	Within vessels
Examples	**HS**, warm AIHA, thalassaemia, sickle cell	**G6PD crisis**, PNH, **mismatched transfusion**, MAHA, malaria, mechanical valves
Splenomegaly	✅	Usually ❌
Hb in plasma/urine	❌	✅ **Haemoglobinaemia, haemoglobinuria**, haemosiderinuria

	Extravascular (90 %)	Intravascular (10 %)
Haptoglobin	↓ (slide pearl)	↓↓
Bilirubin, LDH	↑	↑ (LDH very ↑)

Table D: Haemolytic Crises

	Haemolytic	Aplastic	Megaloblastic
Cause	Infection, drugs	**Parvovirus B19**	Folate/B12 exhaustion
Bilirubin	**↑**	**Unchanged**	—
Retics	↑	**Absent**	↓
Extra	—	—	Pancytopenia, neuro/GI

Table E: HS vs G6PD Deficiency

	Hereditary spherocytosis	G6PD deficiency
Inheritance	**AD**	**X-linked** (males)
Defect	**Membrane** (spectrin, ankyrin)	**Enzyme** (HMP shunt → ↓ NADPH → ↓ GSH)
Haemolysis	Chronic, **extravascular** (spleen)	**Episodic**, triggered, mainly **intravascular**
Film	**Spherocytes**	**Heinz bodies** (supravital), **bite cells**
Index	**↑ MCHC**	Normal between attacks
Test	**Osmotic fragility ↑**; ⭐ **EMA binding**	⭐ **G6PD assay 2 months after the attack**
Treatment	Folate; **splenectomy (> 4 y)** for severe disease	**Avoid oxidants**; transfuse in crisis

Table F: Aplastic Anaemia Causes ("anti's" on the slide)

Category	Examples
Idiopathic (50 %)	T-cell autoimmune
Drugs	**Carbimazole**, **chloramphenicol**, chlorpropamide (slide); *[added]* carbamazepine, gold, sulfonamides, cytotoxics
Chemicals	**Benzene**, insecticides
Radiation	e.g. radioactive phosphorus
Infection	EBV, hepatitis, disseminated TB (*[added]* parvovirus → red cells only)
Immune	SLE
Replacement	Leukaemia, lymphoma (myelophthisis)

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Megaloblastic anaemia → give folate	⚠️ **Never folate alone until B12 deficiency is excluded** (neurology worsens)
B12 deficiency = anaemia	**SCD/dementia can occur without anaemia** *[added]*
↑ homocysteine = B12 deficiency	Also **folate**. **MMA** separates them (↑ only in B12)
B12 deficiency appears quickly	**2–3-year stores** → years; folate → months

Rule	Exception / nuance
Pancytopenia = marrow failure	Also **megaloblastic** and **hypersplenism**
Aplastic → retics low	⭐ **Absent retics = most specific**
Haptoglobin separates intra- from extravascular	❌ **Low in both** (slide 35)
Haemolytic crisis → ↑ jaundice	**Aplastic crisis (parvovirus): pallor ↑, bilirubin unchanged**
G6PD assay in a crisis	❌ **False-normal** (young RBCs); test **2 months later**
G6PD → females spared	Carriers are usually unaffected; *[added]* lyonisation can make some symptomatic
HS = microcytic	*[added]* MCV often normal; **↑ MCHC** is the clue
Splenectomy for HS	Wait until **age > 4**; vaccinate first *[added]*
Thalassaemia/sickle cell (hemoglobinopathies in the file title)	**Not taught in this deck**; only listed

Pattern recognition:

- **alcoholic + macrocytosis + hypersegmented PMNs + ↑ homocysteine, normal MMA** → **folate deficiency**. Neurological signs here = **Wernicke (B1)**, not SCD.
- **vitiligo/thyroid disease + glossitis + ataxia + ↓ vibration** → **pernicious anaemia** → anti-IF antibodies, endoscopy.
- **pancytopenia + no splenomegaly + absent retics + recent carbimazole/chloramphenicol** → **aplastic anaemia**.
- **jaundice + splenomegaly + gallstones + family history + ↑ MCHC** → **HS**.
- **Middle-Eastern male + dark urine after fava beans/primaquine/infection** → **G6PD deficiency**.

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5. Clinical Correlations

- **Dementia work-up** always includes **B12** (slide pearl).
- **Metformin** (diabetes) and long-term **PPIs** → B12 deficiency *[added]*. Check B12 in a diabetic with neuropathy, which may not be diabetic.
- **Methotrexate / trimethoprim / phenytoin** → folate deficiency. Prescribe folate with methotrexate *[added]*.
- **Pregnancy:**
  - **folic acid** for anyone planning or capable of pregnancy (NTD prevention)
  - folate demand ↑ in pregnancy and haemolysis
- **Carbimazole** (Thyroid 1): warn the patient to report **sore throat or fever** → check FBC (agranulocytosis / aplasia).
- **Aplastic patients:**
  - **no IM injections, no shaving with blades**
  - neutropenic fever → **broad-spectrum (imipenem)** ± antifungal/antiviral
- **G6PD deficiency** is common in **Saudi Arabia (Mediterranean variant, severe)**:
  - test before **primaquine/tafenoquine** (Travel Medicine), **dapsone, rasburicase**
  - counsel about **fava beans**
- **Haemolytic anaemia, chronic:**
  - **folic acid** prophylaxis
  - watch for **gallstones**, **parvovirus aplastic crisis**

- **iron overload** with transfusions → **desferrioxamine**

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6. Rapid Revision (Last-Day)

- **B12 physiology: animal only**, 1 µg/day, **IF (parietal cells) + terminal ileum**, transcobalamin II, **liver stores 2–3 years**; DNA + **myelin**.
- **B12 causes:**
  - **pernicious anaemia** (northern Europe, F > M, > 30 y, **HLA-A3/B7**, 3 antibodies: anti-parietal, **blocking**, **binding**; + thyroiditis, vitiligo, Addison's, Graves'; ↑ **gastric cancer**)
  - gastrectomy, ileal disease (Crohn's, TB), ***D. latum***, vegans, TC-II deficiency
- **B12 pathogenesis:** nuclear–cytoplasmic asynchrony → megaloblasts → **ineffective erythropoiesis (↑ LDH, ↑ iron, ↑ muramidase)** → **pancytopenia**.
- **B12 clinical: glossitis**, diarrhoea, HSM, bleeding, infection; **SCD** (spinocerebellar, lateral corticospinal, dorsal columns); **dementia**.
- **B12 labs:**
  - **MCV > 115**, normal MCHC
  - **Howell–Jolly, Cabot rings, hypersegmented PMNs**, megaloblastic marrow
  - serum B12 low (< 200; slide's 200–800 = normal)
  - **↑ MMA + ↑ Hcy = B12**; **↑ Hcy + N MMA = folate**
  - therapeutic retic response
- **Schilling test:** 1000 µg IM then 1 µg oral labelled; normal ≥ 25 %, **< 5 % = malabsorption**; stage 2 + IF corrects in **PA**.
- **B12 treatment: hydroxocobalamin IM** (slide: 100 µg daily → monthly for life); transfuse if Hb < 7 or HF/IHD.
- **Folate:**
  - **vegetables**, 50 µg/day, **3-month stores**
  - causes: **alcohol**, pregnancy, haemolysis, dialysis, malabsorption, **MTX, trimethoprim, anticonvulsants**, cirrhosis
  - **no CNS**; serum folate N 6–20; **FIGLU** ↑ (historical)
  - **folic acid 5–15 mg/day**
  - ⚠️ **never folate alone in B12 deficiency**; folate for anyone who could become pregnant (NTD)
- **Aplastic types:**
  - **unipotent**: red cells (↓ EPO in renal failure; thymoma/oat-cell; parvovirus aplastic crisis), **TAR**, neutropenia
  - **pluripotent** → pancytopenia
- **Aplastic causes: idiopathic 50 % (T-cell)**, radiation, **carbimazole, chlorpropamide, chloramphenicol**, **benzene**, insecticides, TB/EBV/hepatitis, SLE, myelophthisis.
- **Aplastic clinical:** bleeding, **fungal/opportunistic infection without pus**; skeletal deformity (Fanconi).
- **Aplastic labs:** normocytic, **absent retics (most specific)**, **hypoplastic marrow**. DD pancytopenia: hypersplenism, megaloblastic, aplastic.
- **Aplastic treatment:** RBCs, platelets, **no trauma/IM injections**, imipenem + antifungal + antiviral; steroids/androgens/GM-CSF/lithium (slide); immunosuppression; **BMT** (*[added]* ATG + ciclosporin + eltrombopag).
- **Haemolysis physiology:** RBC 120 d; anaemia when 15–20 d; **90 % extravascular**.
- **Haemolysis causes:**
  - **intrinsic**: membrane (**HS**, elliptocytosis, PNH), enzyme (**G6PD, PK**), Hb (thal, sickle)

    - **extrinsic**: mechanical (valves, **MAHA**: DIC, HUS, TTP, malignant HTN, scleroderma), hypersplenism, immune (warm/cold AIHA; transfusion, Rh), infection (malaria, sepsis), toxins (venom, lead, copper)
- **Extravascular features: mongoloid facies**, HSM, **indirect jaundice + pigment stones**, **haemosiderosis** (cardiomyopathy, bronze diabetes, testicular atrophy), urine darkens on standing.
- **Crises: haemolytic** (↑ bilirubin), ⭐ **aplastic = parvovirus (no ↑ bilirubin)**, megaloblastic (folate depletion).
- **Intravascular:** haemoglobinaemia/uria once **haptoglobin**/haemopexin/albumin are exhausted.
- **Haemolysis labs:**
    - **↑ LDH, ↑ indirect bilirubin, ↑ urobilinogen, ↑ retics**, ↑ WBC/platelets
    - **^{99m}Tc** RBC survival ↓
    - **hair-on-end** skull
    - ↑ iron/ferritin
    - **↓ haptoglobin (in both types; don't use it to discriminate)**
- **Haemolysis treatment:** transfusion, **folate**, **desferrioxamine SC**, splenectomy, treat the cause.
- **HS:**
    - **AD**, **spectrin/ankyrin** → microspherocytes
    - family history
    - **osmotic fragility: starts at 0.6 %** (normal 0.4 → 0.33 %)
    - **EMA most sensitive/specific**
    - **splenectomy after age 4** for transfusion need, family history of death, hypersplenism
- **G6PD:**
    - **X-linked**; HMP shunt → NADPH → reduced glutathione
    - deficiency → Hb oxidation → **Heinz bodies** → RES destruction
    - triggers: **infection, antimalarials, anti-TB, sulpha, aspirin, synthetic vit K, DKA, fava beans**
    - **A⁻ mild (old RBCs)** vs **Mediterranean severe (young RBCs)**
    - **methyl violet** Heinz bodies
    - **assay 2 months post-attack**
    - transfuse in crisis; **avoid triggers**

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7. Exam-Style Questions

Q1. (Lecture case) A 44-year-old woman who drinks half a bottle of vodka daily is confused, with **horizontal nystagmus** and a **wide-based gait**. She takes omeprazole and HCTZ. Hb 9 g/dL; **homocysteine ↑, MMA normal**; film shows **hypersegmented neutrophils**. Cause of her **anaemia**? **A. Folate deficiency** · B. Vitamin E deficiency · C. Iron deficiency · D. Vitamin B6 deficiency · E. Vitamin B1 deficiency · F. Vitamin B12 deficiency

Answer: A *[reasoning added; key not on slides]*.

- Hypersegmented PMNs = megaloblastic. **↑ homocysteine + normal MMA = folate** (slide 11). Alcohol is the classic cause (poor intake, ↓ absorption, ↓ hepatic storage).
- **Distractors:**
- the confusion, nystagmus and ataxia are **Wernicke encephalopathy (B1, E)**, but B1 deficiency doesn't cause megaloblastic anaemia. *[added]* Give **thiamine before glucose**.
- **B12 (F)** would raise MMA
- omeprazole can lower B12 and iron, but MMA is normal and the cells are megaloblastic
- **B6 (D)** → sideroblastic, microcytic

Q2. A 60-year-old woman with vitiligo and hypothyroidism has glossitis, paraesthesiae, ↓ vibration sense, positive Romberg and extensor plantars. MCV 118 fL. Best **specific** confirmatory antibody, and one mandatory follow-up? A. Anti-TPO; thyroid US · **B. Anti-intrinsic factor; upper GI endoscopy** · C. ANA; renal biopsy · D. Anti-tTG; colonoscopy · E. Anti-parietal cell; none needed

Answer: B (slides 6–7, 10, 13).

- **Pernicious anaemia with SCD.**
- *[added]* **Anti-IF antibodies are specific**; anti-parietal cell antibodies are sensitive but non-specific.
- **Endoscopy** because of the ↑ **gastric carcinoma** risk.

Q3. A GP gives empirical **folic acid** to a patient with macrocytic anaemia without checking B12. Two months later Hb has improved but **ataxia and numbness are worse**. Why? A. Folate toxicity to nerves · **B. Unrecognised B12 deficiency; folate corrected the blood picture while consuming B12, worsening SCD** · C. Iron deficiency unmasked · D. Hypokalaemia · E. Coincidental MS

Answer: B (slide 18). Always exclude or treat B12 deficiency before folate alone.

Q4. A 25-year-old has pancytopenia 2 months after a course of **chloramphenicol**. No splenomegaly, MCV 92, **reticulocytes 0.1 %**. Most informative test? A. Serum B12 · B. Haptoglobin · **C. Bone-marrow trephine biopsy** · D. Osmotic fragility · E. G6PD assay

Answer: C (slides 21, 23). **Hypocellular (fatty) marrow** confirms aplastic anaemia. Absent retics is the most specific blood finding.

Q5. A child with known HS develops a fever and rash. Pallor worsens, bilirubin is **unchanged** from baseline, and **retics are 0 %**. Cause? A. Haemolytic crisis · **B. Aplastic crisis (parvovirus B19)** · C. Megaloblastic crisis · D. Sequestration crisis · E. Iron deficiency

Answer: B (slide 31). Pallor ↑ without ↑ jaundice or bilirubin = aplastic crisis.

Q6. A 20-year-old Saudi man passes dark urine 2 days after a **fava bean** meal. Film: **bite cells**; supravital stain: **Heinz bodies**. His G6PD level during the episode is **normal**. Next step? A. Diagnosis excluded; investigate PNH · **B. Repeat the G6PD assay ~2 months after the attack** · C. Splenectomy · D. Start primaquine · E. Osmotic fragility test

> **Answer: B** (slide 40). Young RBCs made during the crisis carry normal enzyme → **false-normal** result.

Q7. Which is the **most sensitive and specific** test for hereditary spherocytosis? A. Osmotic fragility · B. Blood film · **C. EMA binding (flow cytometry)** · D. Coombs test · E. Haptoglobin

> **Answer: C** (slide 37).

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Treat megaloblastic anaemia with folate first	❌ **Exclude/treat B12 first**; folate alone worsens SCD
↑ homocysteine = B12	❌ ↑ in **both**; **MMA ↑ only in B12**
Wernicke signs → B12 deficiency	❌ **B1**; the anaemia in an alcoholic is usually **folate**
B12 "200–800 pg/mL" = low (slide 11)	❌ That's the **normal** range; deficiency **< 200**
Megaloblastic marrow has high retics	❌ **Ineffective erythropoiesis → low retics**; retics rise only after treatment
Anti-parietal cell Ab confirms PA	⚠️ Sensitive, not specific; **anti-IF** is specific *[added]*
Pancytopenia → aplastic anaemia	❌ Also **megaloblastic** and **hypersplenism**; **absent retics + hypocellular trephine** for aplasia
Aplastic anaemia with splenomegaly	⚠️ Rethink: leukaemia, myelofibrosis, hypersplenism *[added]*
Chlorpropamide = anti-emetic (slide 21)	❌ **Sulfonylurea**
Aplastic treatment = steroids/lithium (slide)	⚠️ *[added]* **HSCT or ATG + ciclosporin + eltrombopag**
Haptoglobin tells intra from extravascular	❌ **Low in both** (slide 35)
Aplastic crisis → deeper jaundice	❌ **No** rise in bilirubin; **parvovirus**, absent retics
G6PD normal during the attack → excluded	❌ **Retest at 2 months**
Heinz bodies on routine film	❌ Need **supravital (methyl violet)** stain; routine film shows **bite cells**
G6PD affects women equally	❌ **X-linked**: males
HS splenectomy in infancy	❌ **After age 4** (sepsis risk); vaccinate *[added]*
HS is Coombs-positive	❌ **DAT negative** (vs warm AIHA) *[added]*
"Hemoglobinopathies" covered in this deck	❌ **Not taught**; only listed. See the coverage gap note